

$\therefore$ The matrix $\begin{bmatrix} 1 & 0 & 17/8 \\ 0 & 1 & -3/4 \\ 0 & 0 & 0 \end{bmatrix}$ is in reduced row echelon form. $\square$

By defn, Col 1 and 2 have pivots. There are no pivots in Col. 3.

$\therefore$ By defn, x_3 is a free variable.

$\therefore$ The equation $Ax=b$ has a nontrivial solution. $\square$

3. $-3x_1 + 5x_2 - 7x_3 = 0$
 $-6x_1 + 7x_2 + x_3 = 0$

Solution: It's easy to see that $x_1 = x_2 = x_3 = 0$ is a trivial solution of this system of equations. From Ex-1.2, Prob 29, we can say that this system of eqns. has a nontrivial solution since it has infinitely many solutions. $\square$

5. Write the solution set of the given homogeneous system in parametric vector form.

$x_1 + 3x_2 + x_3 = 0$
 $-4x_1 - 9x_2 + 2x_3 = 0$
 $-3x_2 - 6x_3 = 0$

Solution: $x_1 \begin{bmatrix} 1 \\ -4 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ -9 \\ -3 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 2 \\ -6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

By defn. of $Ax=b$, we have:

$\begin{bmatrix} 1 & 3 & 1 \\ -4 & -9 & 2 \\ 0 & -3 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$. From Theorem 3, the ~~matrix~~ ^{system of eqns. to the} augmented matrix $[A \ b]$ has the same solution set as the ^{augmented} matrix $[A \ b]$

Row 1 = Row 1 $\rightarrow$ $\begin{bmatrix} 1 & 3 & 1 & 0 \\ -4 & -9 & 2 & 0 \\ 0 & -3 & -6 & 0 \end{bmatrix}$ $\xrightarrow[\text{+4} \times \text{Row 1}]{\text{Row 2} = \text{Row 2}}$ $\begin{bmatrix} 1 & 3 & 1 & 0 \\ 0 & 3 & 6 & 0 \\ 0 & -3 & -6 & 0 \end{bmatrix}$ $\xrightarrow[\text{+Row 2}]{\text{Row 3} = \text{Row 3}}$ $\begin{bmatrix} 1 & 3 & 1 & 0 \\ 0 & 3 & 6 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Row 1 = Row 1 $\rightarrow$ $\begin{bmatrix} 1 & 0 & -5 & 0 \\ 0 & 3 & 6 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ $\xrightarrow[\times 1/3]{\text{Row 2} = \text{Row 2}}$ $\begin{bmatrix} 1 & 0 & -5 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Verification of whether the final matrix is in reduced row echelon form

1. Row 1 and 2 are non-zero rows. Row 3 is a zero row.

$\therefore$ The condition that all non-zero rows must be above the zero rows is True.